Analytical Report
How has the profitability and R and D intensity of neuroscience company evolved

Final Project Report

Author: Isaac Hui

October 18, 2026

Abstract

The abstract is a very brief summary of the report's contents. It should be about half-a-page long. Somebody unfamiliar with your project should have a good idea of what your work is about by reading the abstract alone.

Originality Avowal

I verify that I am the sole author of this report, except where explicitly stated to the contrary. I grant the right to King's College London to make paper and electronic copies of the submitted work for purposes of marking, plagiarism detection and archival, and to upload a copy of the work to Turnitin or another trusted plagiarism detection service. I confirm this report does not exceed 25,000 words.

Isaac Hui

October 18, 2026

Acknowledgements

It is usual to thank those individuals who have provided particularly useful assistance, technical or otherwise, during
your project. Your supervisor will obviously be pleased to be acknowledged as he or she will have invested quite a lot of time overseeing your progress.

Contents

1 Introduction **2**
1.1 Report Structure 2

2 Background **3**

3 Report Body **4**
3.1 Overview of BioArctic 4
3.2 Market Overview 7
3.3 Partnerships 8

4 Implementation **9**
4.1 Data Analysis 9
4.2 Data Comparison 14

5 Legal, Social, Ethical and Professional Issues **15**
5.1 Section Heading 15

6 Results/Evaluation **16**
6.1 Financial Analysis 16
6.2 R and D Analysis 17
6.3 Overview Analysis 17

7 Conclusion and Future Work **18**
Bibliography 19

A Extra Information **20**

B User Guide **21**
B.1 Instructions 21

C Source Code **22**
C.1 Instructions 22

Chapter 1

Introduction

Biotechnology companies in neuroscience, particularly those targeting neurodegenerative diseases, face unique financial challenges due to their high research and development (R and D) intensity. BioArctic AB, a Swedish biotech leader in Alzheimer's and Parkinson's treatments, exemplifies this dynamic through its strategic pipeline development from 2017 to 2024.

This analytical report examines BioArctic's profitability evolution alongside its R and D spending patterns over this period. Key questions addressed include: How has R and D intensity correlated with net income margins? What financing strategies (cash flows, equity raises) sustained operations amid clinical trial costs? And how do these metrics reveal adaptive responses to biotech market volatility?

1.1 Report Structure

Overview of BioArctic

Overview of the Market

Financial Statement of BioArctic

R and D of BioArctic

Conclusion

Chapter 2

Background

My academic journey in Biomedical Science at King's College London, with a neuroscience major, has centered on understanding neurodegenerative diseases through molecular mechanisms and clinical trial design. During my internship at HKU Medical Laboratory, I gained hands-on experience with experimental neuroscience workflows, which highlighted the inefficiencies in traditional clinical trial modeling and patient stratification.

This project marks my deliberate pivot toward computational biology, using Python to analyze BioArctic's financial and R&D data as a bridge between wet-lab neuroscience and quantitative modeling. By reconstructing profitability trends and milestone impacts from 2017–2024, I demonstrate proficiency in data processing, visualization, and API integration—foundational skills for developing computational models that optimize clinical trial design.

The ultimate goal is to build neural network models that predict trial outcomes and patient response, accelerating therapies like Leqembi for Alzheimer's and Parkinson's. BioArctic's real-world transition from R&D burn to revenue inflection provides the perfect dataset to showcase this interdisciplinary approach.

Chapter 3

Report Body

This analysis explores BioArctic's dual-track strategy of commercial excellence and platform innovation. While the global launch of Leqembi provides the company's current financial foundation, its long-term valuation is increasingly tied to its ability to overcome the Blood-Brain Barrier through its BrainTransporter (BT) technology. This report details BioArctic's market leadership, its strategic partnerships with global giants like Eisai and Novartis, and its transition toward becoming a multi-asset technology leader in the neurodegenerative space.

3.1 Overview of BioArctic

BioArctic AB is a Swedish research-based biopharmaceutical company dedicated to developing disease-modifying treatments for neurodegenerative disorders, primarily Alzheimer's disease (AD), Parkinson's disease (PD), and ALS. Founded on cutting-edge antibody research, the company invented Leqembi® (lecanemab) the world's first anti-amyloid beta protofibril antibody drug proven to slow the progression of the disease and reduce cognitive impairment in early Alzheimer's disease. Leqembi has been developed together with BioArctic's partner Eisai, who are responsible for regulatory interactions and commercialization globally.

3.1.1 Revenue breakdown

Sales of Leqembi

The Sales of Leqembi are primarily located in the Nordic countries and on a global scale they receive royalty from Eisai as this company was crucial for their large scale clinical trials and commercialization. Before late 2022, the drug was still under clinical trials therefore main

revenue comes from previous partnership with AbbVie which was concluded in 2022. The drug was FDA approved and commercialized in 2023 and by achieving these milestones, they received MSEK 590 worth of payment from Eisai as part of their 2007 Agreement and MSEK 10 from Royalty. In 2024, around 90% of BioArctic's revenue comes from the royalty of sales of Leqembi, making royalty from Eisai being an enormous part of their revenue to support their financial foundation

Year	Revenue (kSEK)	Operating_income (kSEK)	Net_income (kSEK)	Operating_cash_flow (kSEK)	investing_cash_flow (kSEK)	financing_cash_flow (kSEK)	cash_and_equivalents (kSEK)	Profit_Margin	Cash_Burn_Rate	R&D_Spending (kSEK)
2017	140706	159750	15157	-135327	-2813	560218	1110367	10.772106377837500	-138140	144593
2018	713970	730229	381602	-200057	-3080	0	917307	53.44790397355630	-203137	348627
2019	281772	296598	88468	327165	-3273	-138506	1112770	31.39701602714250	323892	208130
2020	62347	65943	-68517	-92341	-12524	-6598	999940	-109.8962259611530	-104865	134460
2021	23146	26688	-119789	-140457	-4412	-7389	848405	-517.5365073878860	-144869	146477
2022	228291	229861	-11179	-31638	-12763	-2808	805386	-4.896820286388860	-44401	241040
2023	615995	620077	229249	309694	-507485	14064	611567	37.21604883156520	-197791	390828
2024	257352	257352	-177079	-316332	205633	5686	512927	-68.80809164102090	-110699	434431

3.1.2 Market-Lead Capabilities

R and D Emphasis

Despite having great success in Leqembi with the partnership of Eisai, BioArctic reinvest a lot back in to the research and development of their Brain Transporter technique (BT). And with their partnership with Bristol Myers Squibb and Eisai of BT for Alzheimer's disease being preclinical, they are on task to set this huge milestone of breaking the Blood Brain Barrier Challenge in the whole drug industry. This provide themselves a great foundation for a high valuation of their company especially developing a recent partnership with Novartis focusing on using the BT technique for drug delivery to targets in neurodegeneration.

3.1.3 Key Milestone of BioArcitic's Journey



3.1.4 Strategy in the Pipeline

The milestones achieved between 2017 and 2026 illustrate a deliberate three-phase strategy: Capitalization, Validation, and Platform Expansion.

Phase 1: Capitalization & Infrastructure (2017–2020)

2017 Public Funding: The IPO on Nasdaq Stockholm provided the necessary capital to scale internal operations and retain greater control over proprietary technology.

2019–2020 Final Trials & Recruitment: This period was defined by strategic risk-sharing. By partnering with Eisai for the massive Phase 3 Clarity AD trials, BioArctic offloaded the enormous costs of late-stage clinical development while maintaining significant upside through royalties and milestone payments.

Phase 2: Clinical & Commercial Validation (2022–2024)

2022 Phase 3 Results: The successful readout of the Clarity AD trial served as a historic "proof of concept" for BioArctic's scientific approach to targeting amyloid-beta protofibrils.

2023 Approvals & 2024 Commercialization: With Leqembi® receiving full FDA approval and beginning global rollout, BioArctic transitioned into a revenue-generating entity. In 2024, royalty revenue became the primary driver of the company's financial health, with royalties surging over 180

Phase 3: The "Platform" Strategy (2025–Beyond)

The current strategy leverages the success of Leqembi to fund the BrainTransporter™ (BT) technology, turning BioArctic into a multi-asset technology provider.

Overcoming the Blood-Brain Barrier (BBB): The BT platform is now the core of the company's valuation. By using transferrin receptors to "shuttle" drugs into the brain, it enables up to 70-fold higher brain penetration.

High-Value Partnerships: The 2025 Novartis agreement ($30M upfront, potential $772M milestones) and the BMS/Eisai collaborations validate that the BT platform is not just for Alzheimer's, but is a "delivery engine" for various modalities, including antibodies and genetic medicines.

3.2 Market Overview

3.2.1 Battle of the Alzheimer's Drugs

BioArctic might be the first to discover a FDA approved drug that can slow down the progression of Alzheimer's but a year later

3.2.2 Frontier Innovation competition

3.3 Partnerships

Eisai

Main focus of this project

Bristol Myers Squibb

Novartis

Chapter 4

Implementation

4.1 Data Analysis

4.1.1 Sourcing

All data collected was obtained from the official' company website, ensuring that only publicly disclosed company-verified information was included. I retrieved documents like their annual and quarterly financial statements, investor presentations, and research milestone and pipeline in the R and D section in order for further analysis.

4.1.2 Data Arrangement

Financial Data Assortment for BioArcrtic

Table was created using a Python Package called Pandas. It is a open source library of python that provide data structure and organizations. In this case, Data Frame was used instead of Series for a more efficient and compact data source. By using Data Frame, I was able to construct multiple lists of different variables from 2017 through to 2024 and ultimately building the table listed in Table.

```
import pandas as pd
#Data for BioArctic
BAdata_SEK = {
    "Revenue (SEK)" : [140706,713970,281772,62347,23146,228291,615995,257352],
    "Operating_income (SEK)" :[159750,730229,296598,65943,26688,229861,620077,257352],
    "Net_income (SEK)" : [15157,381602,88468,-68517,-119789,-11179,229249,-177079],
    "Operating_cash_flow (SEK)" : [-135327,-200057,327165,-92341,-140457,-31638,309694,-316332],
    "investing_cash_flow (SEK)" : [-2813,-3080,-3273,-12524,-4412,-12763,-507485,205633],
```

```
    "financing_cash_flow (SEK)" : [560218,0,-138506,-6598,-7389,-2808,14064,5686],
    "cash_and_equivalents (SEK)" : [1110367,917307,1112770,999940,848405,805386,611567,512927]
}

years = [2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024]

BAdf_SEK = pd.DataFrame(BAdata_SEK, index=years)

print(f'BioArctic Financial Data (SEK):\n{BAdf_SEK}')

# Calculate additional financial metrics
BAdf_SEK['Profit_Margin'] = (BAdf_SEK['Net_income (SEK)'] / BAdf_SEK['Revenue (SEK)']) * 100
BAdf_SEK['Cash_Burn_Rate'] = BAdf_SEK['Operating_cash_flow (SEK)'] + BAdf_SEK['investing_cash_flow (SEK)']
BAdf_SEK['R&D_Spending (SEK)'] = BAdf_SEK['Operating_income (SEK)'] - BAdf_SEK['Net_income (SEK)']

metrics_df = BAdf_SEK[['Profit_Margin', 'Cash_Burn_Rate', 'R&D_Spending (SEK)']]

print(f'\nAdditional Metrics:\n{metrics_df}')

#Revenue Growth Rate
revenue_growth = BAdf_SEK['Revenue (SEK)'].pct_change() * 100
print(f'\nRevenue Growth Rate:\n{revenue_growth}')
```

Graph plots

To further analyse these numbers visually, I used the graph plotting function in matplotlab.

```
# Plotting Revenue and Net Income over the years
BAdf_SEK.plot.line( y=['Revenue (kSEK)', 'Net_income (kSEK)'], color=["blue", "red"], title='BioArctic
Revenue and Net Income Over Years')

#Graph plots
fig, axes = plt.subplots(2, 2, figsize=(15, 10))

# Revenue vs R&D Spending
axes[0, 0].plot(BAdf_SEK.index, BAdf_SEK['Revenue (kSEK)'], marker='o', label='Revenue(kSEK)')
axes[0, 0].plot(BAdf_SEK.index, BAdf_SEK['R&D_Spending (kSEK)'], marker='s', label='R&D Spending(kSEK)')
axes[0, 0].set_title('Revenue vs R&D Investment')
axes[0, 0].legend()
axes[0, 0].grid(True)

# Cash flow analysis
axes[0, 1].bar(BAdf_SEK.index, BAdf_SEK['Operating_cash_flow (kSEK)'], label='Operating CF(kSEK)')
axes[0, 1].set_title('Cash Flow Trends(SEK)')
axes[0, 1].legend()

# Profitability
axes[1, 0].plot(BAdf_SEK.index, BAdf_SEK['Profit_Margin'], marker='o', color='green')
```

```
axes[1, 0].set_title('Profit Margin %')
axes[1, 0].axhline(y=0, color='r', linestyle='—')

# Cash position
axes[1, 1].fill_between(BAdf_SEK.index, BAdf_SEK['cash_and_equivalents (kSEK)'], alpha=0.3)
axes[1, 1].plot(BAdf_SEK.index, BAdf_SEK['cash_and_equivalents (kSEK)'], marker='o')
axes[1, 1].set_title('Cash & Equivalents')
```

Currency Conversion

I converted it to a base currency (USD) for comparison by using API. I signed up to a free account on fixer.io to get a private API key for the latest exchange rate. But due to my basic fixer.io plan, it only converts from Euro. To fix this problem, I created a basic formula, which to to divide (the rate of EUR to USD) by (the rate of EUR to SEK) to get the rate of SEK to USD.

```
import requests

def convert_SEK_to_USD_latest(amount_sek,
                              api_key="3d0c0c18a03ae9d790435274ced4d8ad"):
    base_url = "https://data.fixer.io/api/latest"
    params = {
        "access_key": api_key,
        "symbols": "SEK,USD"
    }

    response = requests.get(base_url, params=params)
    data = response.json()

    # Basic error handling, check if the API call succeeded
    if not data.get("success", False):
        raise ValueError(f"API error: {data.get('error')}")

    rates = data.get("rates", {})
    if "USD" not in rates or "SEK" not in rates:
        raise ValueError("Missing USD/SEK in rates data")

    sek_to_usd_rate = rates["USD"] / rates["SEK"]
    amount_usd = amount_sek * sek_to_usd_rate

    rate_date = data.get("date", "latest")
    print(f"Using latest rate from {rate_date}: 1 SEK ~= {sek_to_usd_rate:.4f} USD")
    print(f"{amount_sek:,} SEK ~= {amount_usd:,.2f} USD")

    return amount_usd

if __name__ == "__main__":
```

```
print("Script started")

test_amount = 100000
usd = convert_SEK_to_USD_latest(test_amount,
api_key="3d0c0c18a03ae9d790435274ced4d8ad")
print("Test conversion result:", usd)
```

Whilst programming for the API, I encountered errors initially. My initial plan was to convert each value according to the exchange rate in the same year as the value, however with my free basic plan the endpoint (f"year-12-31") I set came back with multiple quota errors or incomplete historical data. Therefore I decided to just use the latest exchange rate for comparing BioArctic's data to the market.

4.1.3 Milestone Flowchart



Top half: Python + Matplotlib code used to generate the timeline visualization.

Bottom half: Interactive Jupyter Notebook environment (running inside VS Code), which allowed real-time updates, fine-tuning of design, alignment, colors, and text placement while developing the figure.

This combination of static code + live interactive preview was essential for achieving a clear, publication-quality timeline suitable for scientific communication.

Matlab Coding I wrote:

```
import matplotlib.pyplot as plt
import matplotlib.dates as mdates
from datetime import datetime

milestones = [
(2017, "Public Funding", "Successful IPO on Swedish Nasdaq and became the largest biotech listing
since 2000"),
(2019,"Leqembi Final Trials","Continued phase 3 trials with Eisai and preparation for the expansion
```

```
of AD pipeline"),
(2020,"Recruitment","Completed enrollment of 1795 patients in phase 3 study with lecanemab for
Alzheimer's disease"),
(2022,"Results of phase 3","Positive topline results from phase 3, confirming lecanemab's efficacy in
slowing cognitive decline"),
(2023,"Approvals","FDA granted accelerated approval and traditional approval for lecanemab (Leqembi)"),
(2023,"More Approval","Japan (2nd Largest Market for their product in the world) also approved Leqembi"),
(2024,"Marketing and Commercialisation","First sales was in 2023 but with significant commercial rollout
by Eisai in 2024 leads to revenue generation"),
(2024,"Revenue generating strategy","Main income is driven by the royalties from Eisai for lecanemab
sales")
]

# 1. Extract years as datetime objects (perfect for timeline plotting)
years = [datetime(y, 1, 1) for y, _, _ in milestones]

# 2. Extract short titles (these will be the bold/visible event labels)
titles = [title for _, title, _ in milestones]

# 3. Extract detailed descriptions (for hover text, subtext, or smaller font)
descriptions = [desc for _, _, desc in milestones]

# Create figure
fig, ax = plt.subplots(figsize=(12, 10))

# Central vertical line
ax.vlines(0.05, 0, len(milestones)-1, color='lightblue', lw=8, alpha=0.5)

# Plot each milestone (newest at top)
for i, (year, title, desc) in enumerate(zip(years[::-1], titles[::-1], descriptions[::-1])):
    y_pos = len(milestones) - 1 - i  # reverse order: newest on top

    #Horizontal lines underneath the years
    ax.hlines(y_pos, 0.0, 0.19, color='grey', lw=3, alpha=0.5, linestyle='-')

    # Circle marker
    ax.plot(0.05, y_pos, 'o', ms=20, color='royalblue', mec='white', mew=3, zorder=10)

    # Year on left
    ax.text(0.3, y_pos, str(year.year), ha='right', va='center', fontsize=20, fontweight='bold', color=
    'darkblue')

    # Title on right
    ax.text(0.33, y_pos, f"{title}", ha='left', va='bottom', fontsize=15, fontweight='bold',)

    # Description below title
    ax.text(0.33, y_pos-0.15, f"{desc}", ha='left', va='top', fontsize=13,
            bbox=dict(facecolor='lightgrey', edgecolor='lightgray', boxstyle='round,pad=0.6', alpha=0.95))
```

```
# Styling
ax.set_xlim(0, 1.2)
ax.set_ylim(-0.5, len(milestones)-0.6)
ax.axis('off')
plt.title("BioArctic – Lecanemab Development and Commercialization Timeline", fontsize=26, pad=30,
fontweight='bold', color='white', bbox=dict(facecolor='royalblue', edgecolor='none', boxstyle='round,
pad=0.6', alpha=0.7))

plt.show()
```

4.2 Data Comparison

Chapter 5

Legal, Social, Ethical and Professional Issues

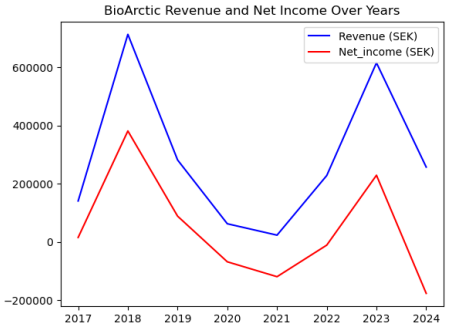
Your report should include a chapter with a reasoned discussion about legal, social ethical and professional issues within the context of your project problem. You should also demonstrate that you are aware of the regulations governing your project area and the Code of Conduct & Code of Good Practice issued by the British Computer Society, and that you have applied their principles, where appropriate, as you carried out your project.

5.1 Section Heading

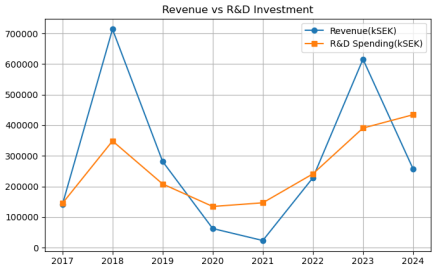
Chapter 6

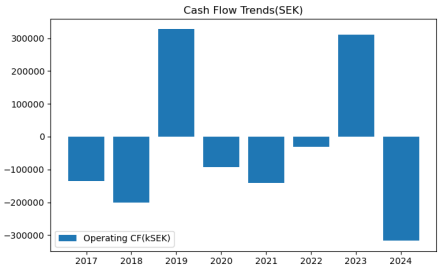
Results/Evaluation

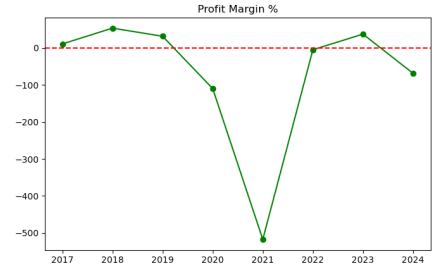
6.1 Financial Analysis

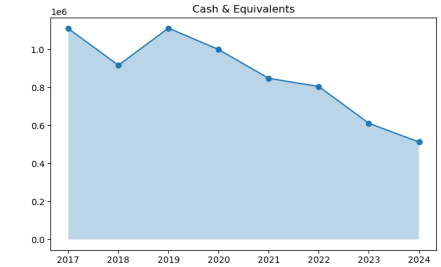


hellooooo









Top Left: Revenue vs R&D investment

blahhblahhbblahhh

Top Right: Clash Flow Trends

Bottom Left: Profit Margin

Bottom Right: Cash and Equivalents

6.2 R and D Analysis

6.3 Overview Analysis

Chapter 7

Conclusion and Future Work

The project's conclusions should list the key things that have been learnt as a consequence of engaging in your project work. For example, "The use of overloading in C++ provides a very elegant mechanism for transparent parallelisation of sequential programs", or "The overheads of linear-time n-body algorithms makes them computationally less efficient than $O(n \log n)$ algorithms for systems with less than 100000 particles". Avoid tedious personal reflections like "I learned a lot about C++ programming...", or "Simulating colliding galaxies can be real fun...". It is common to finish the report by listing ways in which the project can be taken further. This might, for example, be a plan for turning a piece of software or hardware into a marketable product, or a set of ideas for possibly turning your project into an MPhil or PhD.

References

Appendix A

Extra Information

Appendix B

User Guide

B.1 Instructions

You must provide an adequate user guide for your software. The guide should provide easily understood instructions on how to use your software. A particularly useful approach is to treat the user guide as a walk-through of a typical session, or set of sessions, which collectively display all of the features of your package. Technical details of how the package works are rarely required. Keep the guide concise and simple. The extensive use of diagrams, illustrating the package in action, can often be particularly helpful. The user guide is sometimes included as a chapter in the main body of the report, but is often better included in an appendix to the main report.

Appendix C

Source Code

C.1 Instructions

Complete source code listings must be submitted as an appendix to the report. The project source codes are usually spread out over several files/units. You should try to help the reader to navigate through your source code by providing a "table of contents" (titles of these files/units and one line descriptions). The first page of the program listings folder must contain the following statement certifying the work as your own: "I verify that I am the sole author of the programs contained in this folder, except where explicitly stated to the contrary". Your (typed) signature and the date should follow this statement.

All work on programs must stop once the code is submitted to KEATS. You are required to keep safely several copies of this version of the program and you must use one of these copies in the project examination. Your examiners may ask to see the last-modified dates of your program files, and may ask you to demonstrate that the program files you use in the project examination are identical to the program files you have uploaded to KEATS. Any attempt to demonstrate code that is not included in your submitted source listings is an attempt to cheat; any such attempt will be reported to the KCL Misconduct Committee.

You may find it easier to firstly generate a PDF of your source code using a text editor and then merge it to the end of your report. There are many free tools available that allow you to merge PDF files.